New Wheels Project

Introduction to SQL

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Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

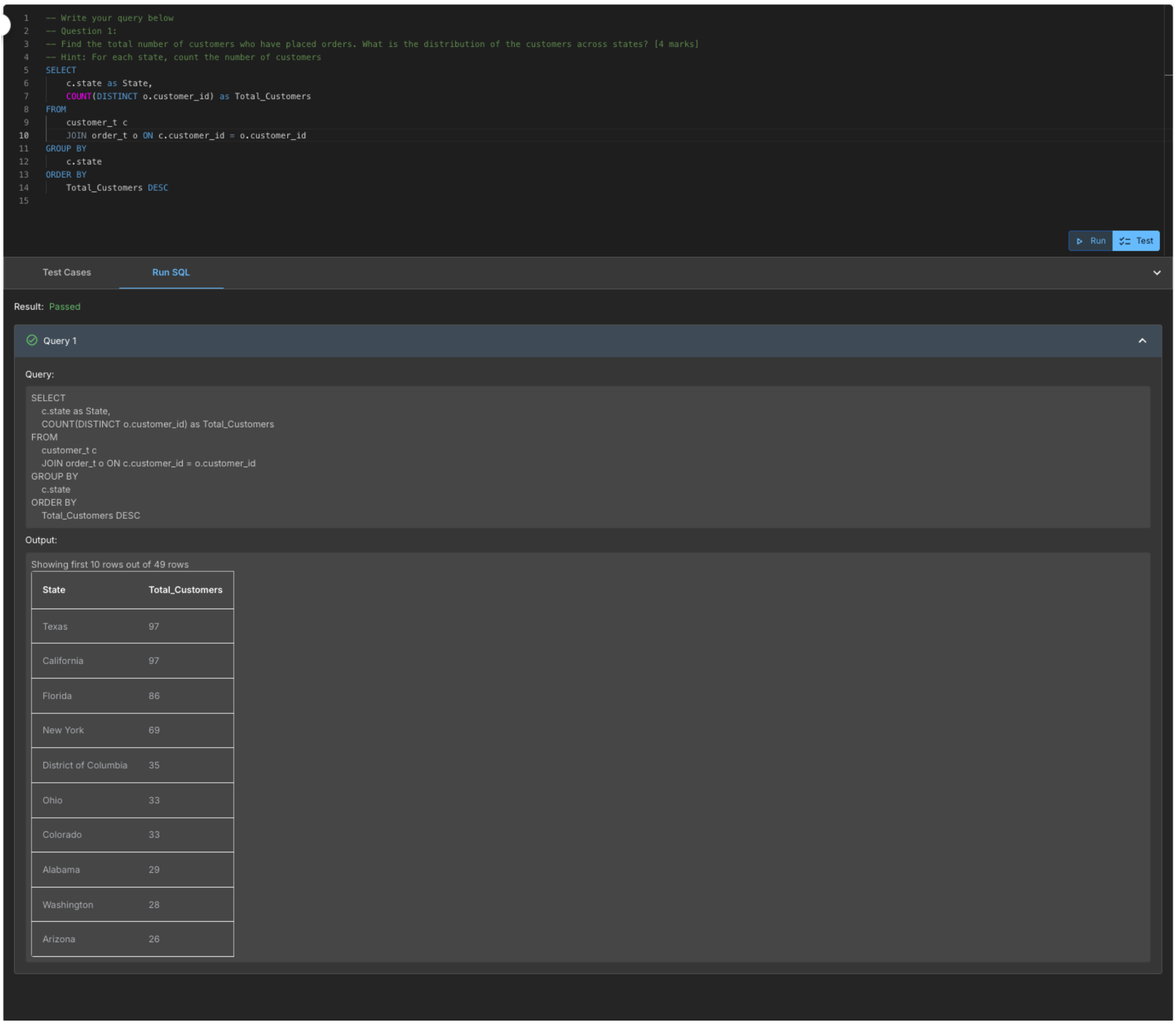
```
SELECT
   c.state as State,
   COUNT(DISTINCT o.customer_id) as Total_Customers
FROM
   customer_t c
   JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY
   c.state
ORDER BY
   Total_Customers DESC;
```

Output:

Observations and Insights:

- The query returns 49 states with a total of 1,000 distinct customers who have placed orders, confirming full national coverage across the dataset.

- Texas and California lead with 97 customers each, followed by Florida (86) and New York (69) — these 4 states alone account for ~35% of the entire customer base, making them the highest-priority markets for New-Wheels.
- Mid-tier states like District of Columbia (35), Ohio (33), Colorado (33), and Alabama (29) represent a significant secondary market that could be grown with targeted campaigns.
- Several states have very low customer counts (single digits), indicating untapped markets — New-Wheels should investigate whether low penetration is due to lack of awareness, logistics constraints, or competitive factors in those regions.



Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker as Vehicle_Maker,
    COUNT(o.customer_id) as Customer_Count
FROM
    product_t p
    JOIN order_t o ON p.product_id = o.product_id
GROUP BY
    p.vehicle_maker
ORDER BY
    Customer_Count DESC
LIMIT 5
```

Output:

```sql
-- ORDER BY
--     Total_Customers DESC

-- Question 2:
-- Which are the top 5 vehicle makers preferred by the customers?

SELECT
    p.vehicle_maker as Vehicle_Maker,
    COUNT(o.customer_id) as Customer_Count
FROM
    product_t p
    JOIN order_t o ON p.product_id = o.product_id
GROUP BY
    p.vehicle_maker
ORDER BY
    Customer_Count DESC
LIMIT 5
```

Run Test

Test Cases Run SQL

Result: Passed

Query 1

Query:

```sql
SELECT
  p.vehicle_maker as Vehicle_Maker,
  COUNT(o.customer_id) as Customer_Count
FROM
  product_t p
  JOIN order_t o ON p.product_id = o.product_id
GROUP BY
  p.vehicle_maker
ORDER BY
  Customer_Count DESC
LIMIT 5
```

Output:

Showing 5 rows

Vehicle_Maker	Customer_Count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

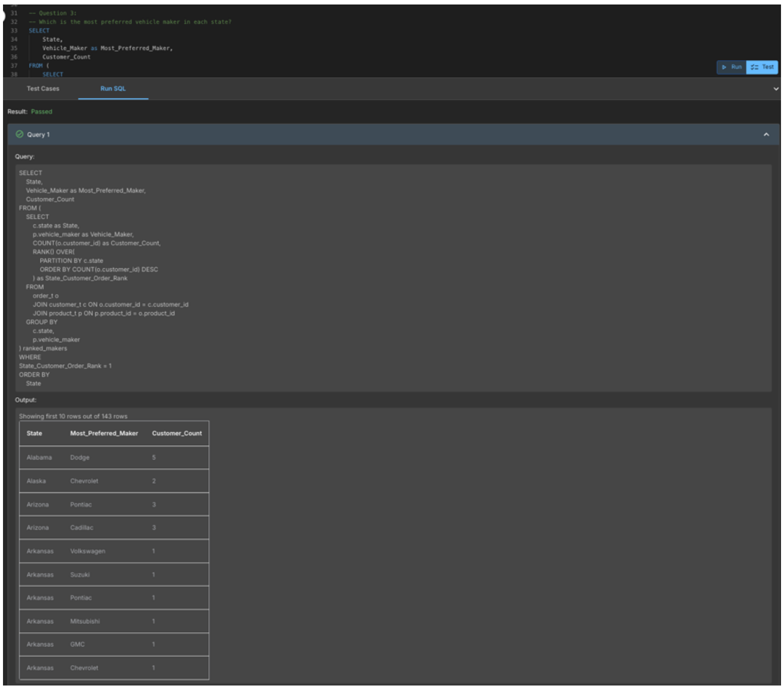
- The top 5 vehicle makers by customer preference are: Chevrolet (83), Ford (63), Toyota (52), Pontiac (50), and Dodge (50) — together accounting for 298 out of 1,000 orders (~30% of total volume).
- Chevrolet leads by a significant margin with 83 orders — 32% more than the second-ranked Ford (63), making it the undisputed most preferred brand and a critical inventory priority for New-Wheels.
- Pontiac and Dodge are tied at 50 orders each, indicating near-equal demand; New-Wheels should maintain balanced stock levels for both brands to avoid stockouts.
- Since these 5 brands drive nearly a third of all orders, New-Wheels should negotiate volume-based procurement deals with these manufacturers to reduce per-unit costs and improve margins.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```sql
SELECT
   State,
   Vehicle_Maker as Most_Preferred_Maker,
   Customer_Count
FROM (
   SELECT
      c.state as State,
      p.vehicle_maker as Vehicle_Maker,
      COUNT(o.customer_id) as Customer_Count,
      RANK() OVER(
         PARTITION BY c.state
         ORDER BY COUNT(o.customer_id) DESC
      ) as State_Customer_Order_Rank
   FROM
      order_t o
      JOIN customer_t c ON o.customer_id = c.customer_id
      JOIN product_t p ON p.product_id = o.product_id
   GROUP BY
      c.state,
      p.vehicle_maker
) ranked_makers
WHERE
State_Customer_Order_Rank = 1
ORDER BY
   State;
```

Output:



Observations and Insights:

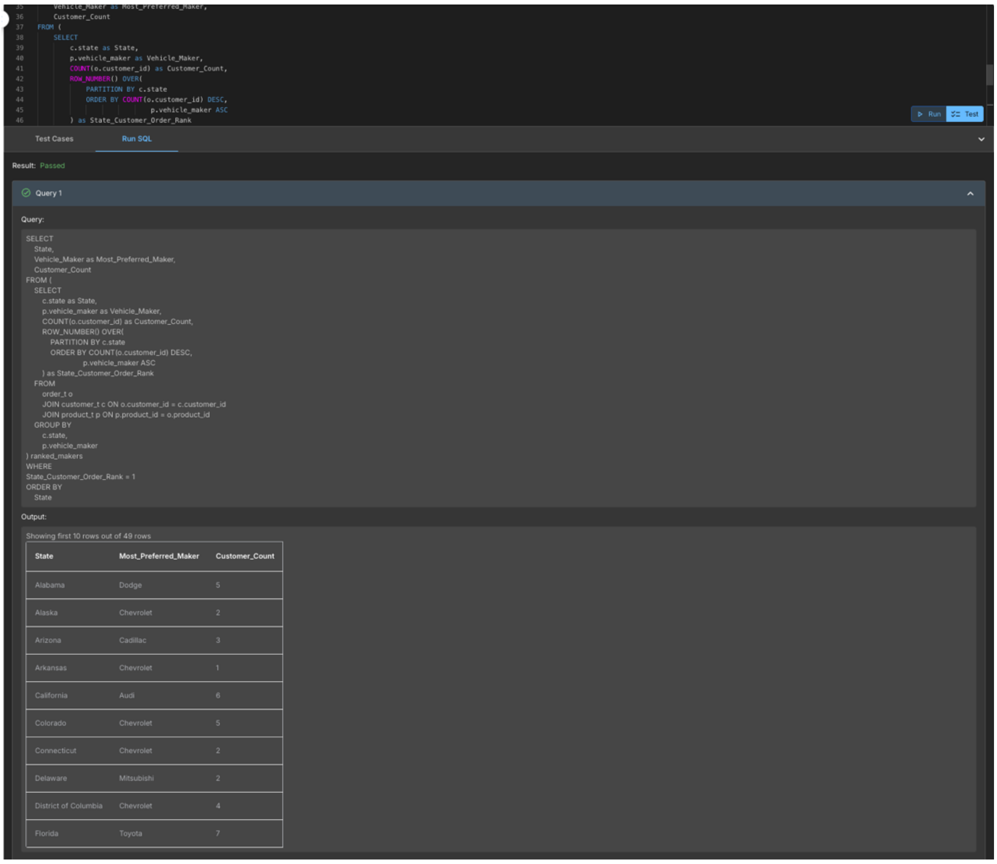
- A window function RANK() is applied partitioned by state and ordered by customer count descending, so rank 1 represents the top maker per state – resulting in rows with ties.
- Since RANK() assigns the same rank to tied values, states like Arkansas (6 makers all at count 1) and Arizona (Pontiac and Cadillac both at 3) return multiple rows; ROW_NUMBER() with a secondary ORDER BY vehicle_maker ASC can be used to return exactly one maker per state.

Alternate Solution Query:

```
SELECT
   State,
   Vehicle_Maker as Most_Preferred_Maker,
   Customer_Count
FROM (
   SELECT
      c.state as State,
      p.vehicle_maker as Vehicle_Maker,
      COUNT(o.customer_id) as Customer_Count,
      RANK() OVER(
         PARTITION BY c.state
         ORDER BY COUNT(o.customer_id) DESC
      ) as State_Customer_Order_Rank
   FROM
      order_t o
      JOIN customer_t c ON o.customer_id = c.customer_id
      JOIN product_t p ON p.product_id = o.product_id
   GROUP BY
      c.state,
      p.vehicle_maker
) ranked_makers
WHERE
State_Customer_Order_Rank = 1
ORDER BY
   State;
```

- Regional preferences for vehicle brands can vary significantly — a brand dominant nationally

Output:



Observations and Insights:

- Chevrolet dominates as the most preferred maker across multiple states including Alaska, Colorado, Connecticut, and District of Columbia — reinforcing its position as the nationally preferred brand.
- California's top maker is Audi (6 orders) — a notable deviation from the national trend — suggesting that premium/luxury vehicles are more popular in high-income West Coast markets, which New-Wheels should account for in regional inventory planning.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Rating Mapping: Very Bad = 1, Bad = 2, Okay = 3, Good = 4, Very Good = 5

Solution Query:

```
SELECT
   'Overall' as Quarter,
   ROUND(AVG(numeric_rating), 2) as Avg_Rating
FROM   (
   SELECT
      CASE customer_feedback
         WHEN  'Very Bad'    THEN 1
         WHEN  'Bad'         THEN 2
         WHEN  'Okay'        THEN 3
         WHEN  'Good'        THEN 4
         WHEN  'Very Good'   THEN 5
      END as numeric_rating
   FROM order_t
) rating_map

UNION ALL

SELECT
   'Q' || quarter_number as Quarter,
   ROUND(AVG(
      CASE customer_feedback
         WHEN  'Very Bad'    THEN 1
         WHEN  'Bad'         THEN 2
         WHEN  'Okay'        THEN 3
         WHEN  'Good'        THEN 4
         WHEN  'Very Good'   THEN 5
      END
   ), 2) as Avg_Rating
FROM order_t
GROUP BY quarter_number
ORDER BY Quarter;
```

Output:

```sql
-- Question 4:
-- Find the overall average rating given by the customers. What is the average rating in each quarter?
-- Rating Mapping: Very Bad = 1, Bad = 2, Okay = 3, Good = 4, Very Good = 5

SELECT
    'Overall' as Quarter,
    ROUND(AVG(numeric_rating), 2) as Avg_Rating
FROM   (
    SELECT
        CASE customer_feedback
            WHEN  'Very Bad'    THEN 1
            WHEN  'Bad'         THEN 2
            WHEN  'Okay'        THEN 3
            WHEN  'Good'        THEN 4
            WHEN  'Very Good'   THEN 5
        END as numeric_rating
    FROM order_t
) rating_map
```

Run

Test

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```sql
SELECT
   'Overall' as Quarter,
   ROUND(AVG(numeric_rating), 2) as Avg_Rating
FROM   (
   SELECT
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         WHEN  'Bad'         THEN 2
         WHEN  'Okay'        THEN 3
         WHEN  'Good'        THEN 4
         WHEN  'Very Good'   THEN 5
      END as numeric_rating
   FROM order_t
) rating_map

UNION ALL

SELECT
   'Q' || quarter_number as Quarter,
   ROUND(AVG(
      CASE customer_feedback
         WHEN  'Very Bad'    THEN 1
         WHEN  'Bad'         THEN 2
         WHEN  'Okay'        THEN 3
         WHEN  'Good'        THEN 4
         WHEN  'Very Good'   THEN 5
      END
   ), 2) as Avg_Rating
FROM order_t
GROUP BY quarter_number
ORDER BY Quarter
```

Output:

Showing 5 rows

Quarter	Avg_Rating
Overall	3.14
Q1	3.55
Q2	3.35
Q3	2.96
Q4	2.4

Observations and Insights:

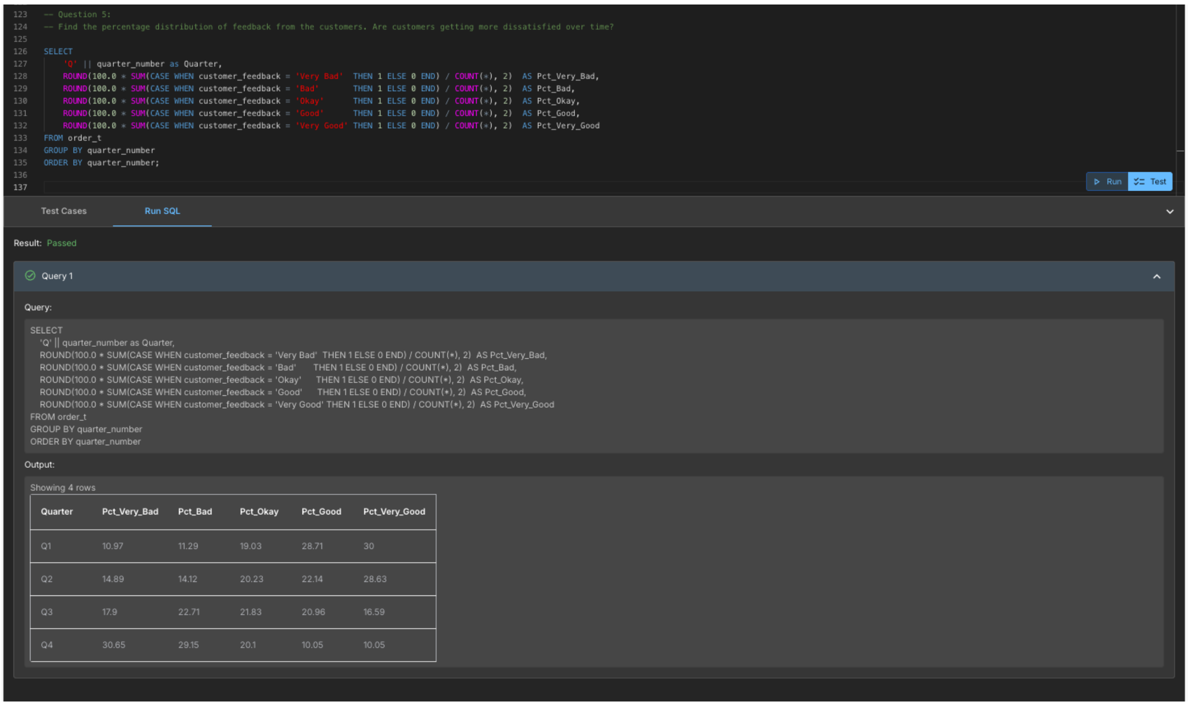
- Text feedback is converted to numeric values using a CASE statement; AVG() is applied both overall and grouped by quarter.
- Quarterly ratings show a clear and consistent decline: Q1 (3.55) and Q2 (3.35) are above average, while Q3 drops to 2.96 — falling below 'Okay' for the first time — and Q4 falls further to 2.40, closer to 'Bad' than 'Okay'.
- The drop from Q1 (3.55) to Q4 (2.40) represents a 32.4% decline in average satisfaction over the year, indicating a progressive and worsening deterioration in customer experience that correlates with the shipping delays observed in Q10.
- Q4's average rating of 2.40 is a critical concern — at this level, the majority of customers rated their experience as 'Bad' or 'Very Bad', which poses a significant risk to customer retention and online reputation if not addressed promptly.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
   'Q' || quarter_number as Quarter,
   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad'  THEN 1 ELSE 0 END) /
COUNT(*), 2)  AS Pct_Very_Bad,
   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad'       THEN 1 ELSE 0 END) /
COUNT(*), 2)  AS Pct_Bad,
   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay'      THEN 1 ELSE 0 END) /
COUNT(*), 2)  AS Pct_Okay,
   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good'      THEN 1 ELSE 0 END) /
COUNT(*), 2)  AS Pct_Good,
   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) /
COUNT(*), 2)  AS Pct_Very_Good
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Observations and Insights:

- Conditional aggregation counts each feedback type per quarter and divides by total orders to yield percentages — the data confirms customers are definitively becoming more dissatisfied over time.
- Combined negative feedback (Very Bad + Bad) rose from 22.26% in Q1 to 59.8% in Q4 — meaning nearly 6 in 10 customers had a bad experience by Q4, up from just 2 in 10 in Q1, representing an alarming deterioration.
- Positive feedback (Good + Very Good) collapsed from 58.71% in Q1 to just 20.1% in Q4 — a 65.8% relative decline — confirming that the business has lost its ability to deliver satisfactory customer experiences as the year progressed.

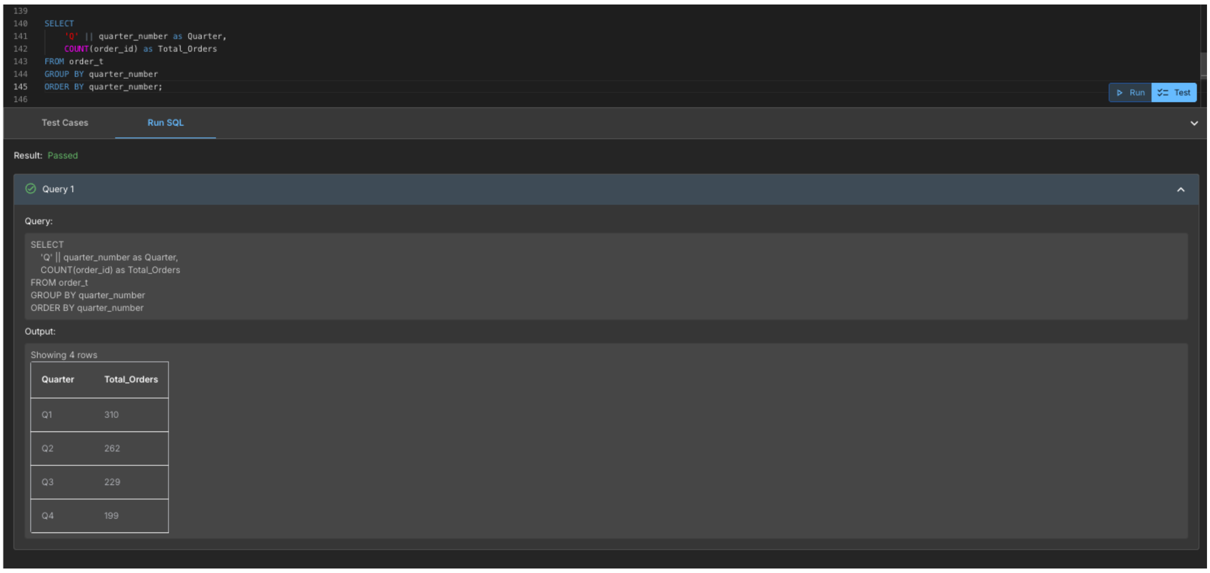
- Q3 marks the inflection point where negative feedback (40.61%) first exceeded positive feedback (37.55%) — this crossover should have triggered an urgent operational review, and New-Wheels must establish real-time feedback monitoring to catch such trends early.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
   'Q' || quarter_number as Quarter,
   COUNT(order_id) as Total_Orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Observations and Insights:

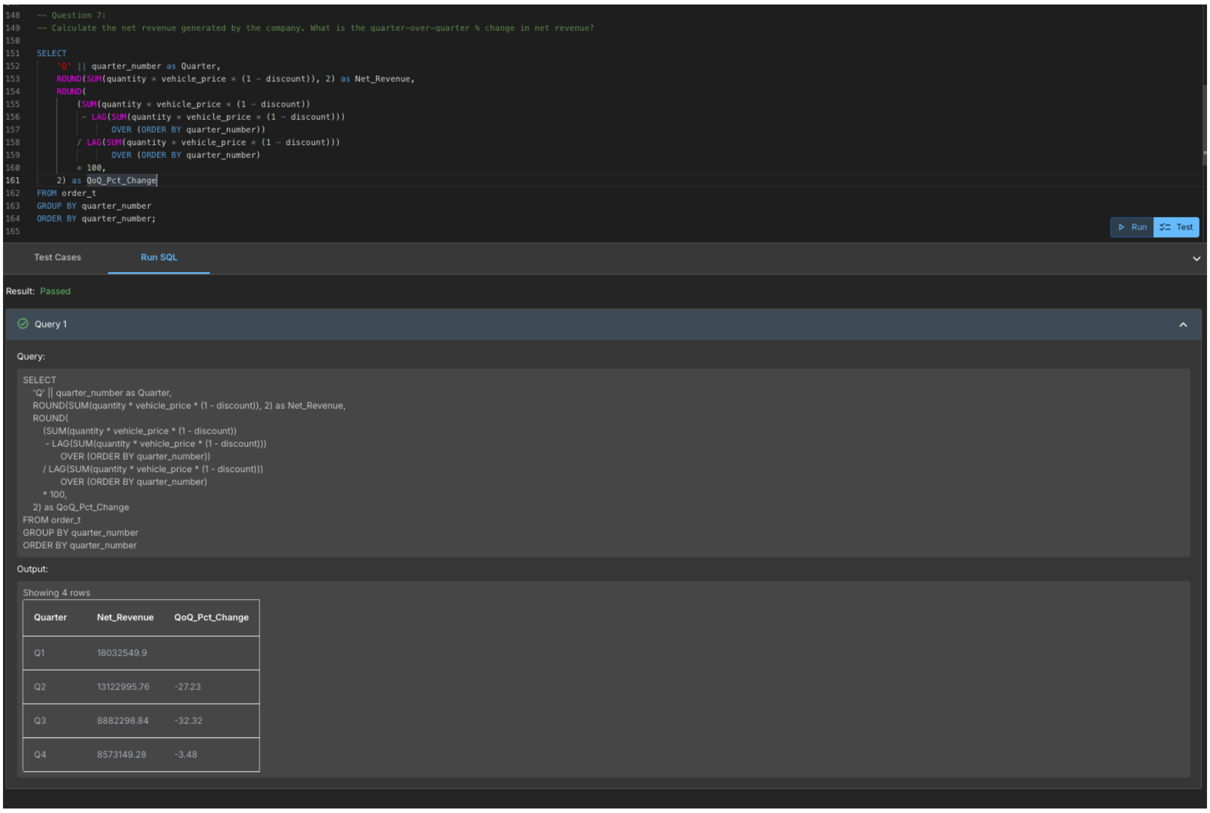
- Orders declined consistently every quarter: Q1 (310) → Q2 (262) → Q3 (229) → Q4 (199), confirming the business concern of falling sales — total orders dropped by 35.8% from Q1 to Q4.
- The steepest drop occurred between Q1 and Q2 (-48 orders, -15.5%), suggesting the customer dissatisfaction that began building in Q1 started driving away repeat buyers and referrals by Q2.
- Q4's 199 orders is the lowest in the year — less than two-thirds of Q1's volume — and directly mirrors the worsening feedback scores and shipping delays, confirming these operational failures are translating into lost business.
- Cross-referencing with Q5, the Q3 feedback crossover (negative > positive for the first time) correlates with the order decline accelerating — this confirms that poor experience is actively deterring new and returning customers.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
   'Q' || quarter_number as Quarter,
   ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) as Net_Revenue,
   ROUND(
      (SUM(quantity * vehicle_price * (1 - discount))
       - LAG(SUM(quantity * vehicle_price * (1 - discount)))
           OVER (ORDER BY quarter_number))
      / LAG(SUM(quantity * vehicle_price * (1 - discount)))
           OVER (ORDER BY quarter_number)
      * 100,
   2) as QoQ_Pct_Change
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Observations and Insights:

- Net revenue is calculated as SUM(quantity * vehicle_price * (1 - discount)); the LAG() window function retrieves the previous quarter's revenue to compute QoQ % change — revenue declined every single quarter: Q1 ($18,032,549.90) → Q2 ($13,122,995.76) → Q3 ($8,882,298.84) → Q4 ($8,573,149.28).

- The largest revenue drop was Q1 to Q2 at -27.23% (-$4.9M), followed by Q2 to Q3 at -32.32% (-$4.2M) — two consecutive quarters of 27-32% revenue decline represent a severe business crisis requiring immediate C-suite intervention.
- Q3 to Q4 showed a smaller decline of -3.48% (-$309K), which may suggest the rate of decline is slowing — however this is likely due to fewer orders rather than any recovery in per-order revenue, as order volume also declined.
- Total annual net revenue is approximately $48.6M, with Q1 alone contributing 37% of that — this heavy front-loading means the business is highly vulnerable to any operational failures that emerge mid-year, and must focus on maintaining Q1-level performance standards throughout the year

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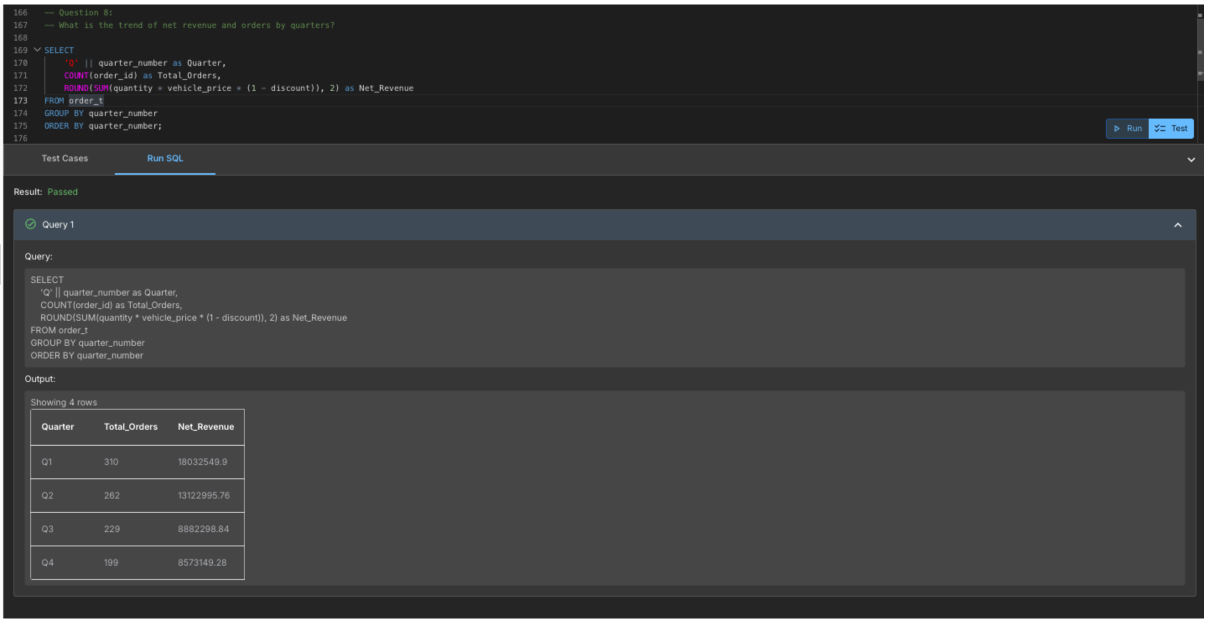
Question 8: What is the trend of net revenue and orders by quarters?Solution Query:

```
SELECT
   'Q' || quarter_number as Quarter,
   COUNT(order_id) as Total_Orders,
   ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) as Net_Revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Observations and Insights:

- Both orders and revenue declined together every quarter — Q1 (310 orders, $18.0M) → Q2 (262, $13.1M) → Q3 (229, $8.9M) → Q4 (199, $8.6M) — confirming this is a broad business health crisis, not just a pricing or discounting issue.
- Revenue per order can be estimated as: Q1 ~$58,170, Q2 ~$50,088, Q3 ~$38,787, Q4 ~$43,081 — the decline in revenue per order from Q1 to Q3 suggests customers may be shifting to lower-priced vehicles or discounts are increasing, compounding the impact of falling order volumes.

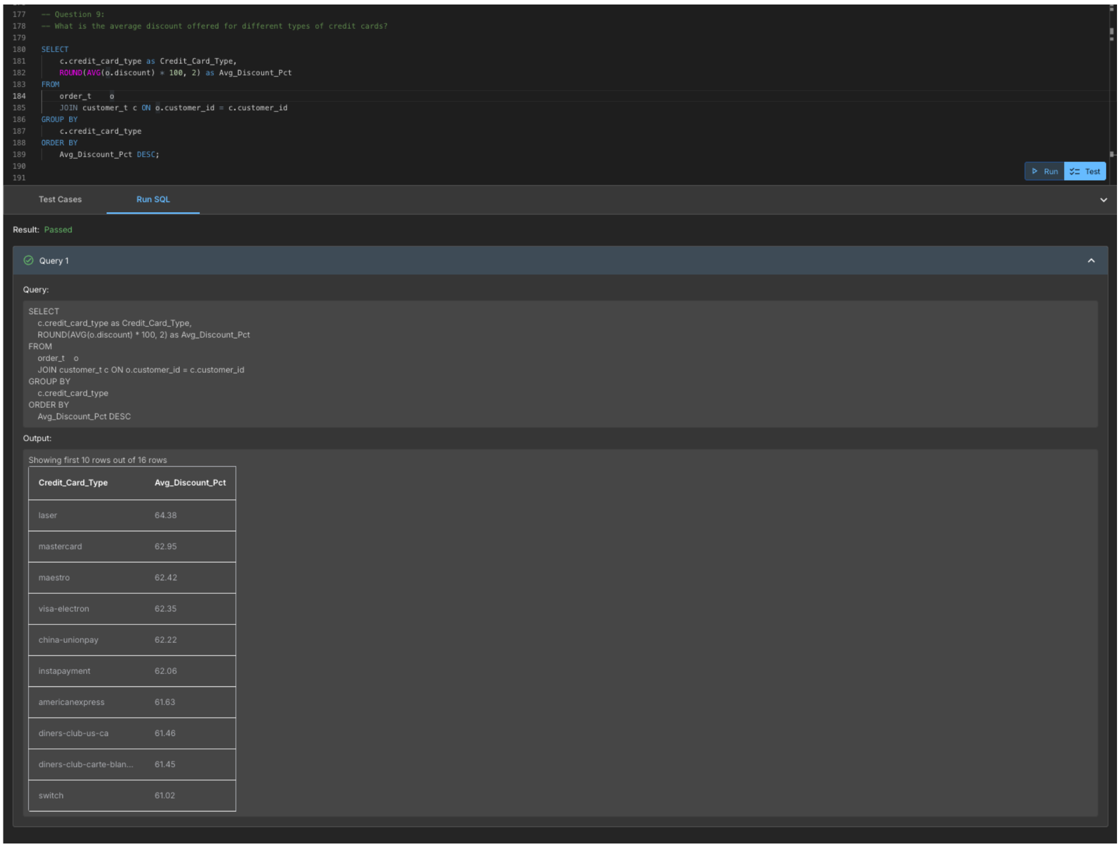
- The slight recovery in revenue per order from Q3 (~$38.8K) to Q4 (~$43.1K) despite fewer orders may indicate that Q4 customers were buying slightly higher-priced vehicles — this is a small positive signal that premium vehicle demand has not entirely collapsed.
- The combined decline in both volume and revenue from Q1 to Q4 means New-Wheels generated 52.4% less revenue in Q4 than in Q1 — restoring Q1-level performance should be the primary business objective for the next financial year.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type as Credit_Card_Type,
    ROUND(AVG(o.discount) * 100, 2) as Avg_Discount_Pct
FROM
    order_t    o
    JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY
    c.credit_card_type
ORDER BY
    Avg_Discount_Pct DESC;
```

Output:



Observations and Insights:

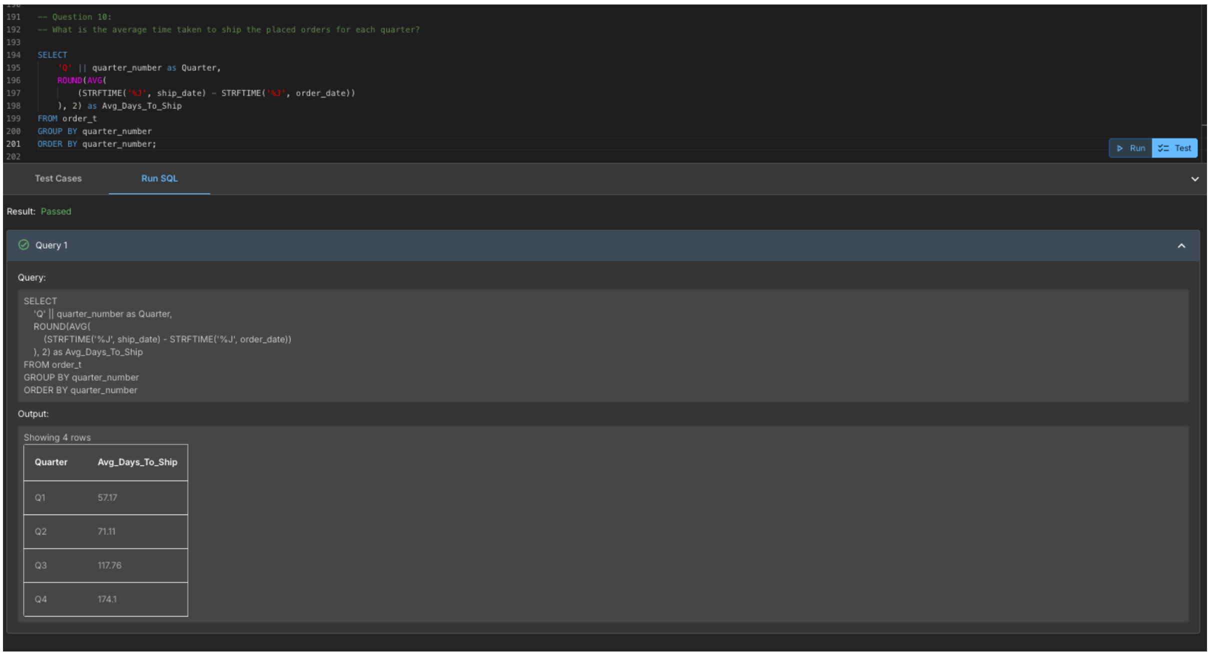
- Joining order_t with customer_t provides credit card type alongside discount; AVG() grouped by credit_card_type shows 16 card types — the top discounts go to Laser (64.38%), Mastercard (62.95%), Maestro (62.42%), and Visa Electron (62.35%).
- The discount range across all 16 card types is very narrow — from 64.38% (Laser) down to ~61% (Switch at 61.02%) — a spread of only ~3.4 percentage points, meaning New-Wheels offers virtually the same discount regardless of card type.
- This near-uniform discount structure indicates that the credit card partnership program is not effectively differentiated — premium or loyalty cards offer no meaningful additional benefit, reducing the incentive for customers to use preferred payment methods.
- New-Wheels should redesign its discount strategy to create clearer tiers — e.g., standard cards at 55%, partner cards at 62%, and premium loyalty cards at 68%+ — to incentivize use of preferred payment channels and potentially negotiate revenue-sharing arrangements with card issuers.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   'Q' || quarter_number as Quarter,
   ROUND(AVG(
      (STRFTIME('%J', ship_date) - STRFTIME('%J', order_date))
   ), 2) as Avg_Days_To_Ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Observations and Insights:

- STRFTIME('%J') is used to compute the difference between ship_date and order_date; the result is averaged per quarter.
- Shipping time increases significantly across quarters — Q1 averages ~57 days, rising to ~174 days by Q4 — indicating a severe and worsening logistics problem over the year.
- The sharp jump from Q2 (71 days) to Q3 (117 days) and Q4 (174 days) suggests a systemic breakdown in the shipping process in the second half of the year, possibly due to rising order volumes overwhelming fulfillment capacity.
- This escalating shipping delay directly correlates with the decline in customer ratings seen in Q3 and Q4 — customers waiting nearly 6 months for delivery are unlikely to give positive feedback, and New-Wheels must urgently restructure its logistics partnerships and warehouse operations to reverse this trend.

Business Metrics Overview

The table below summarises the key performance indicators for the full year based on the query results above.

Total Rewards	Total Orders	Total Customers	Average Rating
$48,610,993.78	1,000	1,000	3.14 / 5
Last Quarter Revenue	Last Quarter Orders	Avg Days to Ship	% Good Feedback
$8,573,149.28	199	104.91 days	20.1% (Q4)

Analysis:

- Total Revenue = Sum of all quarterly net revenues: $18,032,549.90 + $13,122,995.76 + $8,882,298.84 + $8,573,149.28 = $48,610,993.78
- Total Orders = Sum of all quarterly orders: 310 + 262 + 229 + 199 = 1,000
- Total Customers = 1,000 distinct customers across 49 states (Q1 data)
- Average Rating = 3.14 / 5 (Overall from Q4 query)
- Last Quarter = Q4 figures (Revenue: $8,573,149.28 | Orders: 199)
- Avg Days to Ship = Average of all 4 quarters: (57.17 + 71.11 + 117.76 + 174.1) / 4 = 104.91 days
- % Good Feedback = Q4 Pct_Good + Pct_Very_Good = 10.05% + 10.05% = 20.1% (last quarter good feedback)

Business Recommendations

1. Prioritise Improving Shipping and Fulfilment Times

- Average shipping time grew from 57 days in Q1 to 174 days in Q4, a 204% increase over the year. This trend closely parallels the deterioration in customer feedback and order volumes, suggesting that shipping performance may be a key contributing factor to declining business results.
- New-Wheels may consider renegotiating SLAs with logistics partners to target a maximum delivery window of 14–21 days, and introducing performance-based clauses to incentivise faster fulfilment.
- Exploring regional warehousing or last-mile delivery partnerships in the top-volume states (Texas, California, Florida, New York — approximately 35% of the customer base) could help

reduce delivery distances and potentially improve shipping times for a significant portion of orders.

2. Address the Decline in Customer Satisfaction

- Positive feedback fell from 58.71% in Q1 to 20.10% in Q4, while negative feedback rose from 22.26% to 59.80% — a near-complete reversal of sentiment. At this rate, customer retention and new customer acquisition through referrals may be at risk.
- Establishing a structured feedback monitoring process — with defined thresholds that trigger review when negative feedback exceeds a set percentage — may help the business identify and respond to service issues more quickly in future quarters.
- A customer outreach programme targeting Q3 and Q4 customers who reported negative experiences could help New-Wheels understand specific pain points and, where appropriate, offer remediation to rebuild trust.

3. Focus Revenue Recovery in High-Volume States

- Revenue declined from $18.0M in Q1 to $8.6M in Q4 — a 52.4% reduction. Given that Q1 contributed approximately 37% of full-year revenue, sustaining early-year performance levels throughout the year could significantly improve annual results.
- The top 4 states (Texas, California, Florida, New York) account for approximately 35% of the customer base — targeted engagement with existing customers in these states may offer a relatively efficient path to recovering order volumes.
- Reactivating customers who placed orders in Q1 but did not return in later quarters could be a cost-effective approach, as these individuals have already demonstrated willingness to purchase from New-Wheels.

4. Align Inventory and Product Mix with Regional Preferences

- The top 5 vehicle makers — Chevrolet (83), Ford (63), Toyota (52), Pontiac (50), and Dodge (50) — account for approximately 30% of all orders; ensuring consistent availability of these makes may help maintain baseline sales volumes.
- California's most preferred maker (Audi) differs from the national leader (Chevrolet), which may indicate that a single national inventory strategy leaves some regional preferences under-served — state-level stocking decisions informed by Q3 query results could help address this.
- Low-volume states with single-digit customer counts may benefit from a phased approach — starting with a small allocation of locally preferred brands to test demand before committing to larger stock levels.

5. Review the Credit Card Discount Structure

- All 16 credit card types receive very similar average discounts (61.02% to 64.38%) — a spread of only 3.4 percentage points — which means the current programme offers minimal differentiation across card types.
- Introducing a tiered discount structure could create clearer incentives for customers to use preferred payment methods, and may provide a basis for co-marketing arrangements with card issuers that could partially offset discount costs.
- Tracking order conversion rates by card type over time would help assess whether discount differentiation produces a measurable impact on customer behaviour.

6. Introduce a Structured Quarterly Business Review Process

- The consistent quarter-on-quarter deterioration across orders, revenue, ratings, and feedback throughout the year suggests that earlier visibility into these trends — and a structured process for acting on them — may have enabled faster corrective action.
- Establishing defined threshold alerts (e.g., for average rating, QoQ revenue change, shipping time, and negative feedback percentage) and linking them to a regular CEO review cadence could help the business identify and respond to emerging issues more quickly.
- Using Q1 results as a performance baseline and setting quarterly improvement targets for each key metric would provide clear benchmarks to measure progress against and align teams around shared objectives.